Aryan Mittal

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Education

Georgia Institute of Technology Atlanta, GA
Master of Science in Computer Science: Machine Learning, GPA: 4.0 *Expected May 2026*
- *Planned Coursework*: Statistical Learning, Machine Learning Theory, Stochastic Processes, Convex Optimization

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 4.0 *May 2025*
- *Relevant Coursework*: Deep Learning, Machine Learning, Probability & Statistical Theory, Information Theory, Game Theory, Automata & Complexity Theory

Experience

Millennium Management June 2025 — August 2025
Quantitative Developer Intern *Miami, FL*
- Risk Technology Team

Microsoft May 2024 — August 2024
Software Engineering Intern: Azure Core Networking *Atlanta, GA*
- Decreased network node failure detection time from **72 hours** to **1 hour** and saved **8+** maintenance hours weekly by designing and implementing .NET service in C# to continuously monitor the health of **2,000+** routers
- Deployed service to **198** Azure servers **globally** and improved security by eliminating **3,000** yearly remote server logins
- Created utility service to export **16,000** rows of test result data per hour using Kusto queued ingestion API
- Authored comprehensive documentation on both services for use by team of **30+** developers
- **Skills:** C#, .NET Framework, Computer Networking, Azure, Kusto, Azure Data Explorer, PowerShell

Sports Quantitative Researcher January 2023 — Present
Joel Sokol Lab (Georgia Tech) *Atlanta, GA*
- Contracted by **MLB** team to develop optimization model for enhancing league efficiency and competitiveness (confidential, details discussable in person)
- **First-authoring 2 publications** on model and presenting results to MLB Commissioner's Office
- Designed logistic regression Markov chain model to rank NFL teams and predict playoff outcomes with **63%** accuracy (outperforms Vegas spread)
- **Skills:** Python, PyTorch, Scikit-learn, Pandas, NumPy, Gurobi, Linear Programming, Nonlinear Optimization

UPS Supply Chain Solutions June 2023 — August 2023
Software Engineering and Analytics Intern *Alpharetta, GA*
- Prototyped **98%** accurate cloud-based computer vision application for package damage detection and reporting (est. potential annual savings **$5M**)
- Developed **Weibull regression** model to estimate task completion times to within **2** business days
- Created **4** Power BI reports to track work item status across Schedule, Time, and Pay team (**60**+ personnel)
- Corrected pay rates for **350K** drivers by restructuring timecard database (**1.5M** entries) using Python script
- **Skills:** Power BI, Google Cloud Platform, BigQuery, Vertex AI, Python, Pandas, Flask, Java, Azure DevOps

Mathematics Teaching Assistant August 2023 — May 2025
School of Mathematics, Georgia Tech *Atlanta, GA*
- Teach weekly studio/recitation sessions to **70**+ students in Linear Algebra, Multivariable Calculus, and Calculus I

Research and Projects

The Cordiality Game and the Game Cordiality Number | *thearyanmittal.com/projects/cordiality*
- Invented graph-theoretic game with applications in statistical physics and proved **2** optimal play theorems
- Authored final paper **published** in *Graphs & Combinatorics*

Predicting Momentum Swings at Wimbledon | *thearyanmittal.com/projects/mcm*
- Developed **Markov chain** model to capture shifts in competitive advantage across 7000+ Wimbledon plays
- Predicted pivotal points with **41% increase** over baseline F1 score with **XGBoost** and conducted model **ablation study**
- Awarded **Top 20%** (10,000+ teams) in the 2024 COMAP MCM for 100-hour quantitative research challenge

Portal Laptop | *thearyanmittal.com/projects/portal*
- Built **$300** prototype laptop optimized for **20ms** average remote desktop latency with **sub-30 second** user setup time
- Led **178** user interviews, conducted pricing analysis, and performed full statistical evaluation, visualization, and **hypothesis tests** of latency/jitter/packet loss experiments using **mixed effects models**
- Designed and engineered **secure** backend in **Rust** for auth system, peer-to-peer routing, and device management